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Geothermal Energy: Solutions and Future Trends for Managing Environmental and Social Impacts

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Abstract

This paper provides a thorough analysis of the environmental and social risks associated with geothermal energy development. It highlights the interrelationships of these risks and explores ways to sustainably exploit geothermal energy. The paper examines the challenges and limitations of geothermal energy development and proposes solutions, technologies, practices, and policies to mitigate potential environmental impacts, enhance social acceptance, and ensure environmentally responsible production. The research includes a detailed review of scientific literature and case studies from several countries focusing on conventional and enhanced geothermal systems (EGS). The analysis addresses environmental risks, such as induced seismicity, water consumption, and air and water pollution, as well as social acceptance issues stemming from potential risks, trust deficits, and concerns about benefit sharing. The paper will assess the feasibility and scalability of various mitigation technologies, regulatory frameworks, and community engagement strategies.

Geothermal energy is a renewable resource with significant potential, but it also presents environmental and social challenges. A major concern is the induced seismicity associated with EGS, in which fluids are injected into rock formations to fracture and increase their permeability, often triggering microseismicity. Strategies to mitigate this risk include pre-screening sites with detailed geological assessments and using adaptive management techniques that adjust injection parameters based on real-time seismic monitoring. Other strategies include implementing protocols with predefined thresholds for operational adjustments. Water consumption is another important environmental consideration, especially in water-stressed regions. Geothermal power plants can impact water quantity and quality, potentially contaminating aquifers with geothermal fluids or depleting freshwater resources. Solutions include optimizing drilling processes to minimize water use, promoting closed-loop binary cycle power plants that limit water consumption and evaporation, and researching alternative working fluids, such as supercritical CO₂, to reduce water dependence.

Public acceptance is essential for the successful implementation of geothermal projects. Community concerns often stem from perceived risks, limited understanding of geothermal technology, and lack of involvement in decision-making. Strategies to foster social acceptance include engaging with local

communities early and transparently, establishing clear benefit-sharing mechanisms, offering educational programs to address misconceptions, and conducting thorough social impact assessments. Through synthesizing various case studies and conducting an extensive literature review, this research offers a comprehensive framework for sustainably utilizing geothermal resources while minimizing environmental impacts and ensuring social equity and community benefits. Considering the entire life cycle of geothermal projects—from exploration and development to operation and decommissioning—is crucial for mitigating potential risks and enhancing the long-term sustainability of this valuable renewable energy source.

Introduction

Geothermal energy harnesses the Earth's natural heat and stands as a unique, renewable, carbon-neutral energy source. Due to its low carbon emissions and potential to provide secure, baseload power around the clock, it plays an increasingly vital role in the global energy transition and climate change mitigation efforts, unlike intermittent sources like wind and solar. There is huge potential for power generation with the substantial thermal energy reserves within the Earth's crust. However, despite its significant benefits and potential, geothermal energy development is associated with environmental and social risks and challenges. These include geological hazards, such as induced seismicity, which is often linked to techniques like fluid injection in EGS. Other concerns include water usage, potential contamination of water resources, air pollution from non-condensable gases (NCGs), ecological damage, and issues related to land use and waste management. These potential negative effects can raise questions about social acceptability and lead to community opposition. Effectively addressing and managing these impacts is crucial for sustainable development. [Figure-1](#) illustrates the environmental and social risks of geothermal energy development.

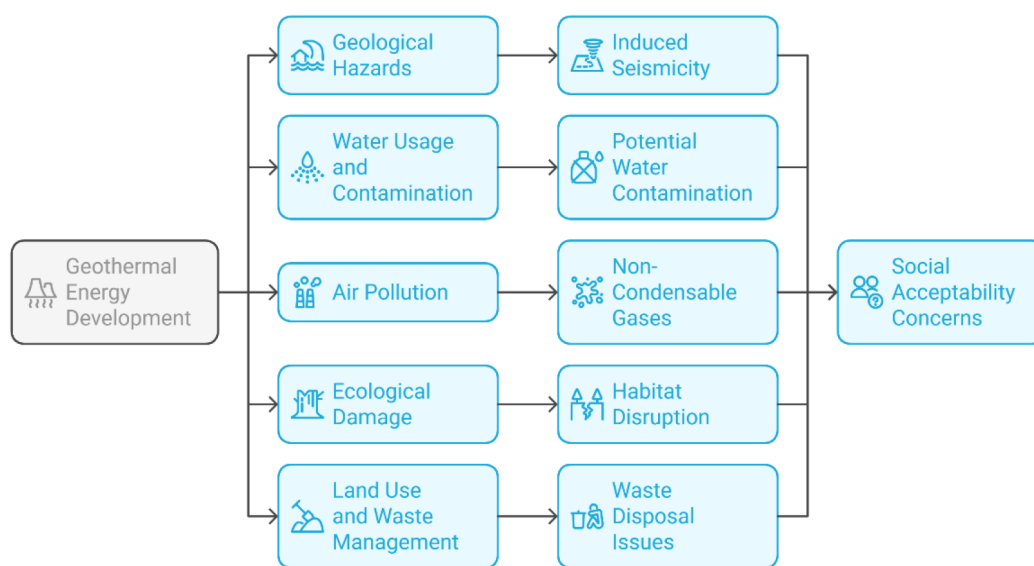


Figure 1—Environmental and Social Risks of Geothermal Energy Development.

Geothermal Energy: Resources and Development Pathways

Geothermal energy utilizes the Earth's heat through various methods tailored to different geological conditions and depths. Developing these diverse resources involves a structured process, or life cycle, that progresses from the initial exploration phase, in which suitable sites are located, to the development and construction of energy conversion facilities, and finally to the long-term operation phase. Realizing the potential of these different types of geothermal energy requires navigating the distinct stages of this project life cycle while ensuring technical feasibility, economic viability, and consideration of environmental and social impacts.

Types of Geothermal Systems

Geothermal systems are categorized based on depth, geological conditions, and extraction methods. *Shallow geothermal systems* are typically found within the upper 400–500 meters of the subsurface. They are commonly exploited using geothermal heat pumps (GHPs), which use the relatively constant ground temperature for heating and cooling up to depths of 100–150 meters. *Conventional geothermal systems*, also known as *hydrothermal systems*, rely on naturally occurring reservoirs of hot water or steam. These reservoirs are often located near tectonic plate boundaries or in volcanic areas, where sufficient heat, fluid, and permeability are present. These systems extract heat from high-permeability rocks containing groundwater, typically at depths between 0.1 and 4.5 kilometers. *Enhanced geothermal systems* target hotter, deeper unconventional resources, such as hot dry rock (HDR), which lack sufficient natural fluid or permeability. EGS uses engineering techniques, such as hydraulic fracturing, to create or enhance subsurface reservoirs and enable fluid circulation to extract heat. EGS typically operates at depths between 2 and 10 km. Beyond these are *advanced geothermal systems* (AGS), which include next-generation technologies such as closed-loop systems and alternative working fluids, like supercritical carbon dioxide (SCCO₂), as well as hybrid systems. These technologies aim to improve efficiency and expand the resource base. Figure-2 summarizes the types, depth range, and potential of conventional geothermal systems and unconventional geothermal development alternatives (Khodayar & Björnsson, 2024).

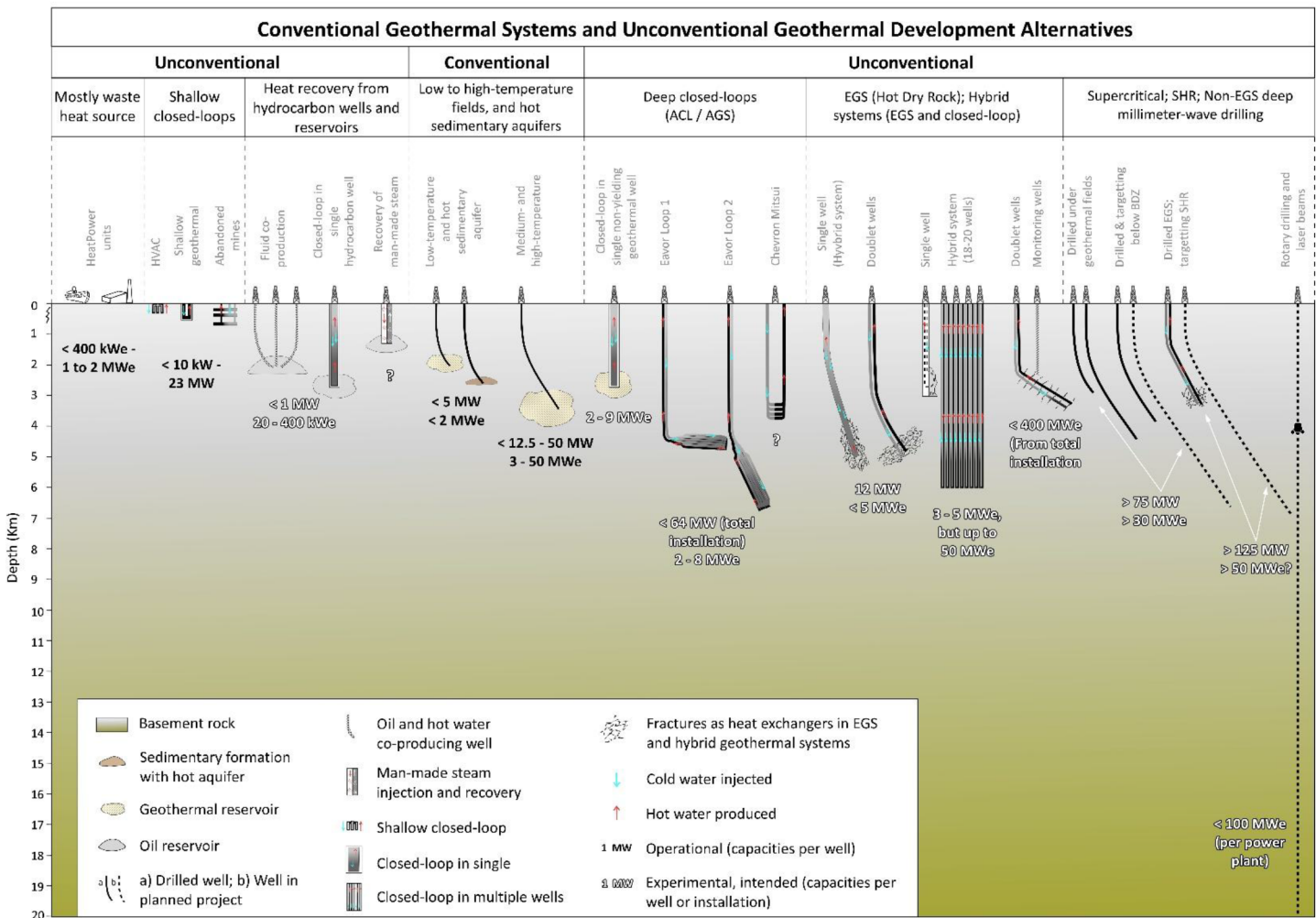


Figure 2—Summary of the conventional and unconventional geothermal systems (Khodayar and Björnsson, 2024).

The Geothermal Project Life Cycle

Developing diverse geothermal resources involves a structured process, or life cycle, that progresses from initial exploration to locate suitable sites, to the development and construction of energy conversion facilities, to the long-term operation phase, which includes ongoing resource management, and finally, decommissioning.

Exploration and Resource Assessment. This is the first step. It involves identifying potential geothermal sites. Key activities include analyzing the geothermal gradient, stress orientation and magnitude, regional tectonics and seismicity, and the geomechanical and hydraulic properties of reservoir rocks, including existing faults, joints, and fractures, as well as mineralogy. This information is gathered through databases and preliminary geological, chemical, and numerical studies. Techniques such as surface manifestation inventory and geological, hydrogeological, geochemical, and geophysical surveys (e.g., seismic and magnetotelluric surveys) are employed to characterize the subsurface. Resource potential can be estimated using methods such as volumetric estimations and numerical modeling.

Development and Construction. Once a viable resource has been identified, wells are drilled to access the geothermal reservoir. Drilling operations require freshwater for drilling mud and other purposes. This phase also involves constructing the necessary infrastructure, including drilling platforms, pipelines, roads, fluid separators, heat exchange stations, and power plants. Planning, financing, and obtaining permits are critical aspects of this stage. Construction activities, including building roads and well pads, can impact soils and increase the project's footprint. Significant upfront capital investment is required, especially for drilling and infrastructure. Figure-3 shows the stages of the geothermal development project and its risk levels (Gehring & Loksha, 2012).

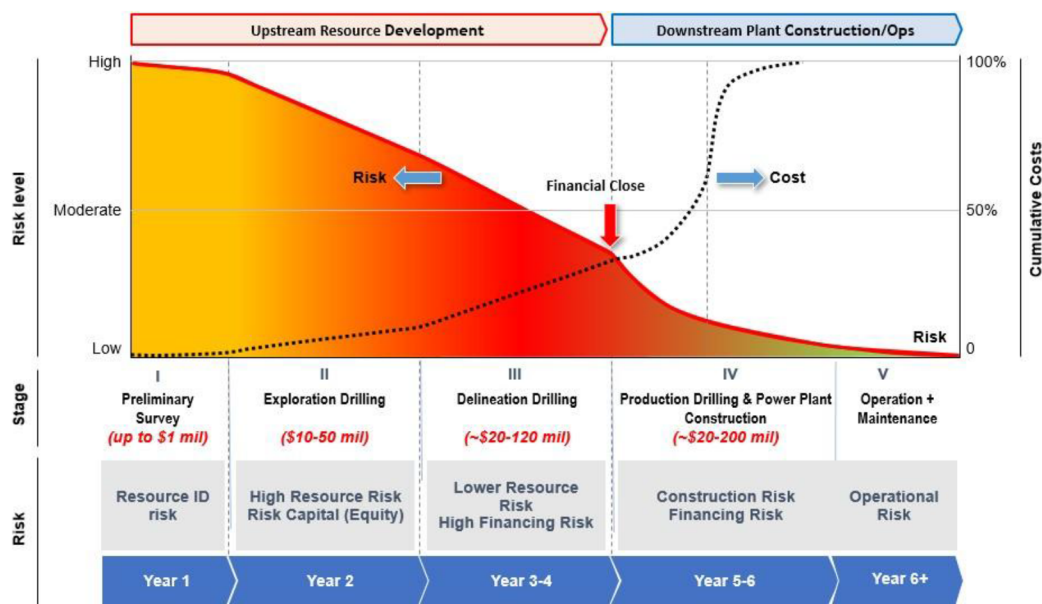


Figure 3—The stages of the geothermal development project and its risk levels (Gehring and Loksha, 2012).

Power Plant Operation. This is the phase in which thermal energy is converted into usable energy for electricity generation or direct applications. For EGS, operational aspects include reservoir stimulation to maintain fluid circulation and optimize energy extraction. Reinjecting spent fluids back into the reservoir is a common practice used to sustain productivity and manage waste. Routine monitoring activities, which can be enhanced by digitalization and data analytics, are performed to track reservoir performance and detect potential issues, such as induced seismicity.

Environmental Impacts of Geothermal Development

Geothermal energy offers significant benefits as a low-carbon renewable resource. However, its development and operation are associated with environmental risks and challenges that require careful management. These impacts are often localized and potentially reversible with the right measures. The specific effects depend on the nature of the geothermal resource, its location, the technology used, and the extent of human activity in the area. Environmental risks can be categorized by their spatial distribution or the part of the environment they affect. Common themes include surface, subsurface, and atmospheric impacts.

Induced Seismicity

One of the most significant environmental concerns associated with geothermal energy development, especially EGS, is the potential for induced seismicity. EGS involves injecting fluids at high pressure into low-permeability hot rock to create or enhance fracture networks for heat extraction. The resulting increase in pore fluid pressure can reactivate pre-existing fault systems and potentially trigger earthquakes. While conventional geothermal operations in tectonically active areas can be associated with an increase in low-magnitude earthquakes (microearthquakes) above the natural background level, the earthquakes generated by geothermal projects are usually microseismic events that are not felt by humans. However, in some cases, EGS operations have induced significant earthquakes. These earthquakes have the potential to cause nuisances, building damage, or even injuries. Induced seismicity can also lead to the early termination of projects, as was seen in Basel, Switzerland, or cause development delays. Although studies indicate that hazardous seismic activity causing major damage has been rare over a century of geothermal deployment, effectively managing seismic risk is crucial for EGS success and social acceptance. Mitigation techniques include careful site selection to understand underlying mechanisms, effective monitoring of operations, managing injection pressure and time (e.g., using cyclic injection or side-leg concepts), and implementing protocols such as the traffic-light system. Understanding seismic hazard potential may require hybrid techniques that combine structural geology and deterministic and dynamic mixed-mode fracture models. [Figure-4](#) illustrates the maximum observed magnitude of seismic events for different EGS projects as a function of the volume of injected fluid ([McGarr, 2014](#)).

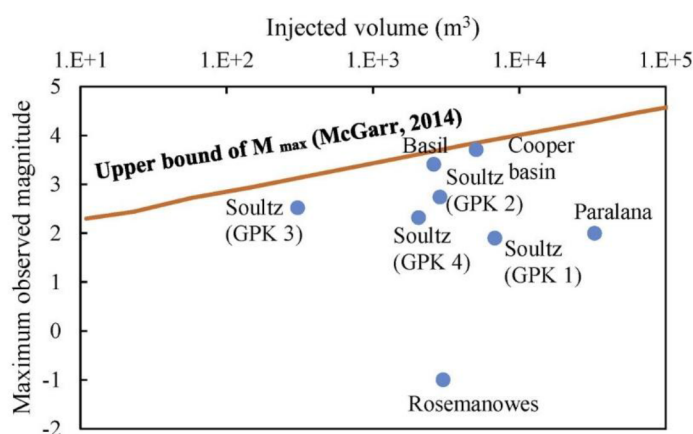


Figure 4—The observed maximum magnitude of seismic events of different EGS projects as functions of volume of injected fluid ([McGarr, 2014](#)).

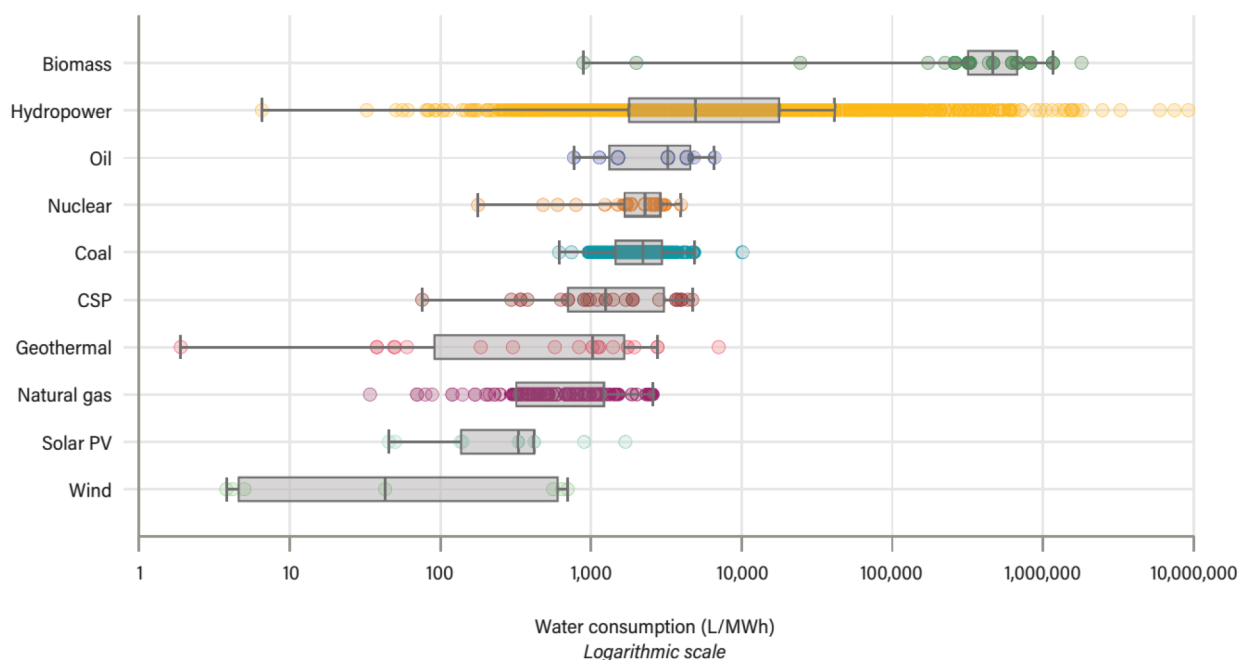
Atmospheric Emissions

Although geothermal energy is generally considered to be a low-emission source, dry steam and flash plants can release non-condensable gases dissolved in geological fluids. These NCGs often consist of carbon dioxide (CO₂), hydrogen sulfide (H₂S), hydrogen, methane (CH₄), helium, argon, oxygen, and

nitrogen. These gases are often released into the atmosphere to maintain operational efficiency. Typical NCG concentrations range from 0.5% to 1.0% of the steam, though this range can be broader (0.2% to 1.8%) depending on the characteristics of the geothermal field. Trace amounts of mercury, arsenic, and radon may also be present. Hydrogen sulfide (H_2S) is a notable concern due to its offensive odor and potential health impacts at higher concentrations. While CO_2 emissions are generally much lower than those from fossil fuel power plants, CO_2 and minor amounts of Sulfur dioxide (SO_2) are still present. Binary power plants, which use a closed-loop system for the geothermal fluid, significantly reduce or eliminate atmospheric emissions compared to flash or dry steam plants. Mitigation techniques include separation and isolation technologies to control emissions, hydrogen sulfide (H_2S) abatement systems (mandatory in some regions, such as the USA), and reinjection of spent brines, which can limit emissions. Geothermal power plants can also generate significant noise, potentially exceeding community standards (e.g., 71–83 dB).

Water Resource Impacts

Geothermal development can affect the quantity and quality of water resources. Significant amounts of freshwater are required during development, especially for drilling mud and dust control activities. During operation, some power plant types, such as flash steam facilities, require freshwater for cooling purposes or to replace water lost through evaporation. Although a global meta-analysis indicates that geothermal energy is less water-intensive than coal or nuclear energy, it is more water-intensive than solar or wind energy. Figure-5 illustrates water consumption over the life cycle of geothermal energy compared to other energy types (Jin et al., 2019). This is a particular concern in water-scarce regions with high geothermal potential, such as the western United States and parts of developing countries. Geothermal operations can also significantly impact water quality.



Notes: CSP = concentrated solar power; L = liters; MWh = megawatt-hour; PV = photovoltaic. Dots represent averages; circles are outliers. Blue water is surface water and groundwater. Geothermal energy includes both conventional geothermal and enhanced geothermal systems (EGS). The median for EGS is 1,670, and the median for conventional geothermal is 244.

Figure 5—Water consumption over the life cycle of geothermal compared to other types of energy (Jin et al., 2019).

Geothermal fluids extracted from deep reservoirs often contain high concentrations of dissolved minerals, chlorides, sulfides, and trace amounts of toxic chemicals, including arsenic, boron, cadmium, lead, mercury,

and fluoride. Radionuclides, such as radium-226 and radium-228, may also be present in mineral scales. If not managed properly, these fluids can contaminate groundwater and surface water through accidental spills, leaks, or inadequate disposal. Contamination of groundwater quality and drinking water production poses a significant threat. Single-well extraction systems that release pumped water onto the surface can lead to a decline in groundwater levels. Reinjecting spent fluids back into the reservoir is critical for managing geothermal fluids. This practice helps sustain reservoir pressure, mitigating subsidence, and prevents surface or shallow groundwater contamination. However, reinjection is also directly linked to induced seismicity. Geothermal wastewater, including extracted geothermal water and drilling fluids, can cause thermal pollution when discharged into surface water bodies. This increases water temperature and harms aquatic life.

Land Use and Habitat Alteration

Geothermal energy operations require land above ground for drilling platforms, well pads, pipelines, roads, fluid separators, heat exchange stations, and power plants. However, geothermal power plants use relatively less land than some other energy sources, such as coal or nuclear power plants. Estimates for land use by geothermal power plants range from about 1 to 8 acres (4 to 32 Km²) per megawatt produced, depending on the infrastructure footprint (De Jesus, 2025). Although geothermal power plants require less land overall, they are often located in natural habitats, sometimes near national parks or tourist areas, which can lead to conflicts regarding land use. Construction activities, such as building roads and well pads, increase the project's immediate footprint. These activities can lead to soil impacts, such as compaction, which reduces aeration and permeability and increases runoff and erosion. Soil warming near geothermal manifestations or infrastructure can also occur, impacting plant life. Site clearing, road construction, and drilling activities may cause habitat disturbance, fragmentation, and a reduction in habitat quality. These changes can lead to the loss of native vegetation and biodiversity, altered water and seed dispersal, and impacts on wildlife due to noise and disturbance. In some cases, geothermal development has been linked to ecological changes, such as alterations in plant cover in warmer streams or the attraction of exotic species that affect native wildlife (Wetterer, 1998). Minimizing habitat loss and implementing appropriate conservation and mitigation measures are essential. Directional drilling can reduce the land footprint by drilling multiple wells from one pad. Some locations practice the concept of multi-use management, where geothermal areas are shared with compatible activities, such as farming or forestry. Forestry projects can enhance groundwater recharge, benefiting the geothermal system and local communities.

Waste Generation

Geothermal energy projects produce various types of waste throughout their lifecycle. During the drilling phase, waste products include drill cuttings and used drilling fluids. These fluids are usually clay-based, such as bentonite, and contain additives that can be stored in ponds. These fluids and cuttings can contain trace elements, sulfides, and naturally occurring radioactive material (NORM). During operation, minerals such as sulfur, silica, and carbonate can accumulate in pipes, turbines, and other equipment as scale. Sludges containing these materials, along with heavy metals such as arsenic, mercury, vanadium, and nickel, may be classified as hazardous waste. Corrosion of equipment can also contribute to waste solids. Compared to other thermal power plants, such as coal or nuclear plants, geothermal plants generally produce less solid waste. Proper waste disposal, including the handling of toxic and hazardous waste by accredited facilities, is necessary. Practices like the "3 Rs" (reduce, reuse, recycle) are encouraged. Reinjection is also a key method for managing spent fluids containing dissolved substances and reducing the volume of surface waste. Figure-6 illustrates the environmental risks of conventional geothermal energy projects (De Gregorio et al., 2020).

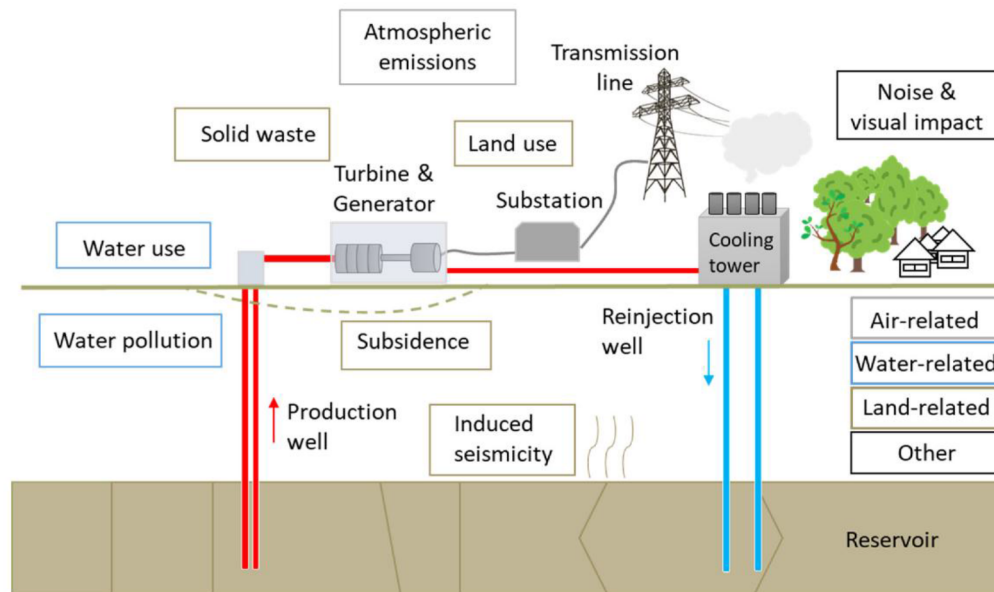


Figure 6—The environmental risks of conventional geothermal energy projects (De Gregorio et al., 2020).

Extraction of Minerals from Geothermal Brines

Although not strictly an environmental impact of development, the chemical composition of geothermal brines raises potential environmental concerns if not managed properly. However, it also presents opportunities for resource extraction. Geothermal fluids can be rich in various elements, including hazardous substances such as arsenic, boron, cadmium, and lead, as well as valuable critical metals like lithium, cesium, and rubidium. Processes like combined heat power and metal extraction (CHPM) integrate energy production with mineral recovery from geothermal fluids (Szanyi et al., 2023). Extracting these valuable metals could enhance the economic viability of geothermal projects and contribute to a circular economy. However, the presence of these elements, particularly the hazardous ones, in geothermal fluids highlights the importance of containment and responsible management practices, such as reinjection, to prevent environmental pollution.

Social Impacts of Geothermal Development

While geothermal energy development offers significant potential as a renewable energy source, it also presents a complex interplay of environmental and social challenges. Successfully implementing and ensuring the long-term sustainability of geothermal projects requires understanding and managing these social impacts and risks.

Community Acceptance and Public Perception

Public acceptance is critical to the successful development of geothermal energy because the success of natural resource projects depends largely on cultural and social acceptability, which is shaped by public perception. In particular, community acceptance hinges on factors such as trust, procedural and distributional justice, and the extent to which the local population supports a project. Despite its potential, geothermal energy development often encounters significant community opposition worldwide. Influential factors include the level of community involvement, anticipated social and environmental impacts, and the perceived distribution of benefits. Fair benefit-sharing is essential for acceptance.

Perceived risks, especially those related to induced seismicity, are central to public debates and can complicate project implementation, as individuals often assess hazards based on perceived rather than actual effects. Community concerns frequently arise from limited understanding of geothermal

technology, insufficient participation in decision-making, and the spread of misinformation-challenges that are particularly pronounced in projects involving EGS, where uncertainty is high. Addressing these issues requires understanding the origins of misinformation and implementing effective debunking strategies. Negative perceptions of environmental impacts and doubts about the adequacy of mitigation measures have led to opposition in various countries, with shortcomings in Environmental and Social Impact Assessment (ESIA) reports exacerbating acceptance problems. These reports should aim to enhance community awareness and address exaggerated concerns. The disparity between perceived positive and negative outcomes can result in locally unwanted land uses scenarios, manifesting as local opposition. To foster social acceptance, it is crucial to engage early and transparently with local communities, establish clear mechanisms for sharing benefits, provide educational programs to dispel misconceptions, and conduct comprehensive social impact assessments. Communication should be interactive and aimed at building trust and a sense of community among stakeholders, as projects that neglect early and inclusive engagement with local populations are more likely to face strong resistance.

Public Health and Safety

Geothermal energy development has minimal implications for local human health compared to other energy sources, such as fossil fuels, biomass, and nuclear power, which are associated with severe health threats. Although geothermal energy typically has moderate environmental impacts, it also poses specific health risks. These risks include exposure to non-condensable gases, such as carbon dioxide and hydrogen sulfide, as well as trace elements, including mercury, arsenic, and radon, which are found in geothermal steam. Some health effects, such as lung cancer linked to geothermal gas exposure, may only become apparent after long incubation periods. Additional risks arise from the potential contamination of groundwater and drinking water due to the injection of chemical additives and operation of geothermal wells. Safety concerns also extend to induced seismicity, particularly with EGS. This necessitates robust management practices, such as seismic risk pre-screening, hazard evaluation, traffic light protocols, continuous seismic monitoring, and comprehensive communication plans. Effectively addressing these health and safety concerns requires mitigating environmental impacts and adopting responsible development practices, including implementing stricter regulatory controls. [Figure-7](#) illustrates a decision tree with a sequence of questions designed to enhance public engagement in geothermal projects (Isabel Fernández Fuentes et al., 2022).

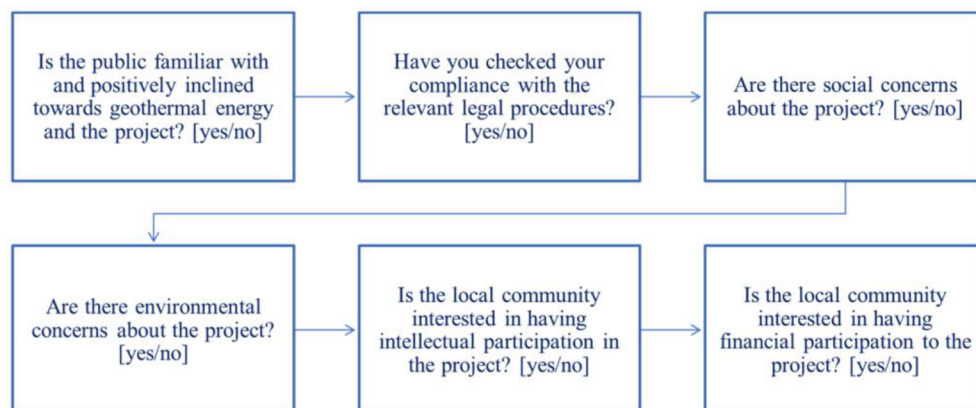


Figure 7—Decision tree's sequence of questions to enhance public engagement (Isabel Fernández Fuentes et al., 2022).

Impacts on Cultural Heritage and Sacred Sites

Geothermal development can significantly impact a region's culture, particularly affecting the lives and heritage of Indigenous Peoples (IPs). These impacts often relate to the cultural ecosystem services that the land provides, including spiritual enrichment, aesthetic value, inspiration, heritage value, nonuse value, and

recreational value. Specific concerns include desecrating ancestral sites and experiencing adverse effects from noise, dust, visual changes, and habitat destruction during construction. These effects can disrupt traditional tribal ways of life and threaten religious and cultural sites. The greatest concern is the potential disruption to IPs' established way of life, including encroachment on ancestral lands and, in some cases, the need for community resettlement. If resettlement is unavoidable, prior and adequate consultation is crucial to reach agreements on relocation, asset replacement, and compensation. Developers must actively support resettled households to help them regain their previous living standards. The United Nations Declaration on the Rights of Indigenous Peoples emphasizes IPs' rights to self-determination, collective ownership of land and resources, and the right to define their own development priorities. When geothermal resources are located on IP lands, adhering to the principle of "free and prior informed consent" is essential. To effectively address cultural impacts, developers should conduct ethnographic studies, respectfully engage with communities, and establish a baseline inventory of sacred sites to avoid disturbing them. Mitigation measures for negative impacts on cultural and religious practices are vital for community acceptance. Ongoing collaboration and engagement with indigenous groups throughout the project lifecycle—including involving IP community members in assessing benefits and risks and integrating traditional knowledge—are key to ensuring cultural concerns are addressed, and project planning and operations are successful.

Solutions, Practices, and Policies for Sustainable Development

Widespread and sustainable geothermal development requires a multifaceted approach that integrates technological advancements, sound practices and frameworks, and supportive policies and regulations. Innovation and collaboration across various domains are required to address environmental and social challenges.

Technological Advancements

Technological innovation is essential to unlocking the full potential of geothermal energy and mitigating its risks. Improved exploration techniques are essential for reducing uncertainty and risk when locating viable geothermal resources. Geophysical imaging techniques, such as seismic surveys, magnetotellurics, and microseismic monitoring, improve reservoir characterization by providing detailed information about subsurface structures, fluid pathways, and reservoir properties. These technologies facilitate improved resource evaluation and management, which can enhance reservoir performance and optimize energy extraction. Understanding the geological controls on geothermal resources is an evolving field. Significant technological developments are needed and are underway for EGS. Improvements in drilling technologies, including directional and slim hole drilling, can reduce costs and minimize environmental impact, making deeper geothermal resources more accessible. Innovative techniques, such as combined thermo-mechanical drilling, are being explored to improve performance in deep, brittle rock formations. Further research and development of drilling and stimulation technologies are necessary to access deeper, hotter resources and reduce overall costs. Technologies for integrated reservoir management are essential for maintaining the long-term performance and sustainability of geothermal systems. This includes optimizing fluid extraction and reinjection processes to sustain productivity and manage waste fluids.

There is also innovation in using geothermal resources for mineral extraction. There is potential for the integration of geothermal energy systems with the extraction of critical metals from geothermal brines. This approach enables the recovery of valuable critical elements, such as lithium and tungsten, and transforms the dual crises of energy scarcity and resource depletion into an opportunity for a circular economy. This methodology is far less ecologically detrimental than conventional mining techniques because it eliminates invasive excavation and mitigates adverse effects on land quality, water resources, and local biodiversity. Combined heat and mineral production technology can unlock new resources and increase the supply of previously untapped ones.

Specific mitigation technologies for environmental impacts are also essential. Technologies are available to manage atmospheric emissions by separating, isolating, and controlling concentrations of certain emissions to acceptable levels. In some regions, H₂S abatement systems are mandatory, and absorption and stripping techniques are available for its removal. Reinjection of spent brines is a key mitigation measure that limits emissions, enhances resilience against depletion, and avoids pollution of waterways. Using waste heat from geothermal plants for direct heating applications increases efficiency and eliminates its release into the atmosphere and waterways. Management measures for solid wastes and precipitates (such as sulfur, silica, and carbonate sludge) collected from plant components include the 3Rs and proper waste disposal. Sludges and scales can be encapsulated in cement to render them inert, converted into soluble forms prior to reinjection, or treated by bacteria.

Practices and Frameworks

In addition to technology, establishing robust practices and frameworks is essential for ensuring sustainable geothermal development throughout the project life cycle. A critical component is implementing comprehensive Environmental Management Plans (EMPs) based on thorough Environmental Impact Assessments (EIAs). The EIA should be conducted before development begins, and the EMP should detail site-specific practices, preventive measures, and mitigation strategies for the entire project. These plans are crucial for obtaining public acceptance and necessary permits. ESIA reports are essential for project funding and implementation, and they should involve public participation to share information and collect opinions from affected communities. An effective EMP ensures that best practices are applied, consent conditions are implemented, environmental laws are complied with, and environmental risks are properly managed.

Clearly, there is a need to refine sustainability assessment frameworks tailored specifically for geothermal projects. Generic frameworks often lack coverage of geothermal-specific issues or rely solely on qualitative assessments. Due to the unique local circumstances of each geothermal project, a customized framework is necessary. This framework should aim to maximize positive impacts, avoid negative impacts, and incorporate sustainability indicators as a measurement tool. Although existing frameworks often rely on qualitative assessment, there is potential for quantitative indicators. Developing a comprehensive, geothermal-specific framework requires extensive, multidisciplinary, international stakeholder consultation to produce a well-rounded set of sustainability goals and indicators. Collaboration among industry, academia, and research organizations is necessary to improve technologies and practices. Multidisciplinary, innovative research involving both the private and public sectors is necessary to address existing restrictions. Although there are general energy frameworks (such as those from the International Atomic Energy Agency or the Commission for Sustainable Development) and frameworks for other renewables (such as hydropower), geothermal projects have unique local circumstances and issues that require geothermal-specific indicator frameworks.

Promoting open access to relevant data through data-sharing platforms is crucial for enhancing the accuracy and reliability of assessment frameworks, as well as for transparency and verification. Limited data availability hinders analysis and reduces the reliability of studies. Therefore, establishing a geothermal data-sharing platform is advised to promote transparency and enable better analysis and verification. Effective and transparent communication and stakeholder engagement strategies are paramount for achieving public acceptance. To encourage broad acceptance, policymakers need to address policy and regulatory concerns. Obtaining public acceptance requires more reliable and understandable communication, as well as public involvement in decision-making processes. Awareness campaigns can promote open communication about the benefits and sustainability of geothermal energy. Comprehensive environmental impact evaluations and stakeholder participation are essential for addressing community concerns. Engaging stakeholders and collaborating with local communities, indigenous groups, environmental organizations, and other relevant parties is necessary to address concerns, obtain support, and ensure proper planning and operation.

Communication should be a two-way process that informs and educates the public about the probability and severity of potential impacts.

Fair and accessible resolution mechanisms are essential for building trust and maintaining harmonious relations with host communities. This can be achieved by addressing social issues, supporting communities, and equitably sharing economic values through corporate social responsibility programs. It is also important to ensure that benefits such as job creation and infrastructure improvement are equitably shared. Respecting cultural heritage, sacred sites, and indigenous rights is critical to sustainable development. Geothermal projects can potentially impact the way of life of indigenous peoples, their ancestral lands, and sacred sites. A baseline inventory of sacred sites should be conducted to avoid such impacts. Project developers should protect the environment, which often overlaps with the ancestral domains of indigenous peoples. This fosters complementary efforts in area protection.

Policy and Regulatory Recommendations

Supportive policies and regulatory frameworks are essential for facilitating sustainable geothermal development. There is a need for coherent regulatory frameworks that address conflicts with other subsurface resource uses. A lack of specific laws regarding spatial development and land use planning can lead to such conflicts. Clear, supportive frameworks are needed to encourage investment, streamline permitting processes, and facilitate project development. To encourage broad acceptance and adoption, policymakers need to address numerous policy and regulatory concerns. Standards must be harmonized, particularly regarding aspects such as geothermal resource inventories and reinjection criteria, as a lack of harmonized standards can be limiting. Some applications, like shallow geothermal energy, require clear guidelines in the legal framework.

Integrating sustainability assessments and EMPs into project approval processes is essential. EMPs should be required for approval or consent, ensuring that preventive strategies and mitigating measures are embedded in the project's design and plans from the beginning. Policies should promote data transparency and public access to information because a lack of publicly available data limits analysis and reliability. A geothermal data-sharing platform can be established through policy. Policies promoting the use of sustainable materials and recycling are needed to support practices like the 3Rs and circularity assessments. Additionally, supportive regulatory frameworks are needed to encourage waste management and material considerations throughout the project lifecycle.

Finally, government support for research and development (R&D) of mitigation technologies and sustainable practices is essential. Policymakers can promote R&D initiatives by funding programs that support technological advancements, pilot projects, and collaborations between research institutions and industries. Investing in R&D fosters innovation and leads to more efficient, cost-effective geothermal technologies. Promoting risk mitigation mechanisms, such as guarantees and insurance, can boost investor confidence and mitigate risks associated with geothermal projects. Stable long-term policies and monetary incentives encourage private investment and market expansion. Setting ambitious renewable energy targets that allocate a significant role to geothermal energy and regularly updating policies is crucial. International collaborations can accelerate technological deployment by sharing best practices and overcoming common problems.

Case Studies and Illustrative Examples

One significant area of focus is community acceptance, which is crucial for the successful realization of geothermal projects. The Olkaria IV Geothermal Project in Kenya is a notable case study that highlights the factors that influence community acceptance. Despite being a leading producer of geothermal energy, Kenya faced significant controversy with the Olkaria IV project. The local Maasai community and non-governmental organizations (NGOs) raised issues including inadequate communication, non-

involvement in decision-making processes, unsatisfactory relocation logistics, and unfulfilled compensation promises. There were also concerns related to impacts on land rights, livelihoods, cultural heritage, and the environment. Third-party mediation processes were conducted to help resolve these disputes. A study of the Olkaria IV project indicated that effectively mitigating environmental impacts, engaging the community in decision-making processes, and clearly communicating socioeconomic benefits can significantly enhance community acceptance (Abdi et al., 2024). The study also emphasized the importance of understanding cultural backgrounds and ensuring the inclusive participation of women and youth in decision-making processes. Although the project's Environmental and Social Impact Assessment (ESIA) reports and Resettlement Action Plans were intended to address potential issues, perceived gaps in these reports contributed to community concerns.

Community opposition, often referred to as "Not In My Back Yard", is a common challenge for geothermal projects in many locations. Concerns that influence public perception in other regions include a lack of trust, spiritual connections to sites (e.g., Hawaii), effects on hot springs (e.g., Japan), association with earthquakes (e.g., Switzerland), negative impacts on cultural heritage, the economy, social life, and the environment (e.g., Indonesia), and visual impacts, gas emissions, and water pollution (e.g., Italy). Successful environmental and social initiatives have been implemented in Philippine geothermal projects, transforming potential conflicts into coexistence. One example is the Northern Negros Geothermal Project (De Jesus, 2010).

Finally, managing seismic risk is a critical challenge, especially for enhanced geothermal systems, where fluid injection can induce seismic activity. Some projects have faced termination or delays due to induced seismicity; examples include the Basel and Pohang projects. Conversely, the Soultz-sous-Forêts project in France, the Helsinki project in Finland, the Blue Mountain project in Nevada, and the Utah FORGE project in the United States are examples of projects where induced seismic risk has been adequately managed. Guidelines for good practices in managing induced seismic risk in EGS projects include pre-screening for seismic risk, conducting comprehensive risk analyses, designing traffic light protocols, monitoring seismic activity, and creating detailed communication and outreach plans tailored to the local social context. These guidelines are based on experiences in geothermal activities and analogous ones, such as hydraulic fracturing in the oil and gas industry.

Conclusion

In summary, while geothermal energy presents a promising source of clean, renewable power, its development is accompanied by significant environmental and social challenges. Key environmental risks include induced seismicity, especially with enhanced geothermal systems; the release of non-condensable gases, such as carbon dioxide and hydrogen sulfide; potential contamination of water and soil by geothermal fluids containing trace metals and brines, if not properly managed; and impacts on land use and biodiversity. These environmental impacts are often closely linked to social concerns and issues of community acceptance. Public perception and social responsibility challenges involve addressing a lack of trust, impacts on land rights and cultural heritage, ensuring equitable benefit sharing, and overcoming rejection sentiments.

Despite the challenges, geothermal energy has significant potential as a sustainable, clean, and reliable baseload energy source. Geothermal energy plays a crucial role in global decarbonization efforts and in ensuring a secure energy supply. However, realizing this potential fully depends on its responsible development. To achieve sustainable geothermal utilization, it is necessary to implement integrated solutions, robust practices, and supportive policies throughout the entire project life cycle, from exploration to decommissioning. This includes conducting comprehensive environmental and social impact assessments; ensuring effective stakeholder engagement and transparent communication; and proactively identifying and mitigating risks through state-of-the-art seismic monitoring and adaptive management protocols. Comprehensive frameworks that integrate environmental protection, social responsibility

(including equity and community benefits), and economic viability are essential for guiding sustainable geothermal development.

Future Research Directions

To maximize the potential of geothermal energy as a sustainable resource, continued and targeted research and development are essential. Key areas requiring further investigation span technical, environmental, social, and governance domains, including the following:

Reducing Exploration Uncertainty

Research is needed to develop improved exploration data and technologies, as well as advanced exploration measures, to reduce technical and financial risks in the early stages of projects. More accurate methods for assessing geothermal potential are also needed.

Enhancing Long-Term Performance Prediction

Understanding the long-term performance and sustainability of geothermal systems is crucial, including fluid flow in complex fracture systems and the long-term thermal and chemical attributes of reservoir rock under extreme conditions. Accurate modeling of reservoir variations and coupled thermo-hydro-mechanical and chemical processes requires sophisticated, data-intensive numerical modeling. The role of Artificial Intelligence (AI) and machine learning (ML) in data processing and predictive modeling should be enhanced.

Improving Mitigation Strategies

Induced Seismicity. Continued refinement of EGS methods is ongoing. Research on the mechanisms and mitigation techniques of induced seismicity is needed, with a focus on increasing permeability without causing negative seismic responses. Robust monitoring, adaptive management, and communication plans are critical areas for further study and implementation.

Fluid and Emissions Management. Research is needed on sustainable mitigation of elements like arsenic in geothermal fluids. Exploring economic methods for extracting high-value metals, such as lithium, from geothermal brines is a significant research opportunity. Continued focus is also required on effective waste reduction and disposal practices.

Developing More Sophisticated Sustainability Assessment Tools

Currently, a comprehensive and systematic assessment framework specifically for geothermal development is lacking. Future research should focus on creating clear definitions and quantifications of environmental impacts and developing multidisciplinary quantitative assessment methodologies. These methodologies could leverage data-sharing platforms and integrated valuation tools, including those for cultural ecosystem services.

Addressing Regulatory Gaps and Harmonization

Research is needed to inform supportive regulatory frameworks and clear policies for land use and permitting. Exploring effective international collaboration and sharing best practices can help harmonize regulations across jurisdictions.

Effectively Managing Social Dynamics and Enhancing Community Acceptance

Understanding the factors that drive community acceptance and addressing trust deficits and public perception are key research areas. Further research is needed on ensuring the equitable distribution of benefits and enhancing community inclusion. Research into effective stakeholder engagement and communication in diverse cultural and socioeconomic contexts remains vital.

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